

SCHOOL OF COMPUTER SCIENCE

What Does Your Social Media Preference Say About You? An Investigation of User Traits and Interface Interaction

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Abstract

The usability of an interface is a significant consideration during the development of any system, since it describes the cognitive compatibility between a system's structure and the user's mindset. It is vital to ensure that the interaction is not only efficient but also tailored to an individual's parameters. In this study, the relationship between the personality of the user and their affinity for a specific user interface (UI) is examined, in the realm of social media. Three mock interfaces were created that represent three different paradigms within social media system design philosophy. Participants were then exposed to these interfaces and given simple tasks to complete to simulate authentic interactions. They were subsequently surveyed on their impression of the interface and finally a measure of their own personality. The system optimised for Agreeability and Conscientiousness had a mean positive emotional state score of 2.7 for users aligned with these traits, compared to 2.1 for non-aligned users, showing that personality-congruent design fosters somewhat higher emotional engagement. Overall, a user's Openness predicts their satisfaction to a moderate extent (r=0.4) with using any interface, indicating how such users are generally more receptive to diverse and novel digital environments. There further exist some relationships between preference for the mock interfaces and social media platform of a similar type, giving validity to the idea that different social media utilise bespoke interface elements to target users of different dispositions. Such results can be used to gain deeper insight into the interplay between a user and the platform. They can be used to facilitate the design of more bespoke interfaces that increase subjective satisfaction, alongside promoting more ethical development by identifying users uniquely susceptible to "dark patterns" in design.

Dedication and Acknowledgements

To my supervisor, Professor Matthew Edwards, for his invaluable supervision and support.

Declaration

I declare that the work in this dissertation was carried out in accordance with the requirements of the University's Regulations and Code of Practice for Taught Programmes and that it has not been submitted for any other academic award. Except where indicated by specific reference in the text, this work is my own work. Work done in collaboration with, or with the assistance of others including AI methods, is indicated as such. I have identified all material in this dissertation which is not my own work through appropriate referencing and acknowledgement. Where I have quoted or otherwise incorporated material which is the work of others, I have included the source in the references. Any views expressed in the dissertation, other than referenced material, are those of the author.

Vojtech Martinat, Tuesday 13th October, 2026

AI Declaration

I declare that any and all AI usage within the project has been recorded and noted within Appendix A or within the main body of the text itself. This includes (but is not limited to) usage of text generation methods incl. LLMs, text summarisation methods, or image generation methods.

I understand that failing to divulge use of AI within my work counts as contract cheating and can result in a zero mark for the dissertation or even requiring me to withdraw from the University.

Vojtech Martinat, Tuesday 13th October, 2026

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Ethics Statement

- An ethics application for this project was reviewed and approved by the faculty research ethics committee as application 29035. A survey was performed as a part of this study. Total number of participants was 40.

Supporting Technologies

- The survey hosting website used was Qualtrics [66].
- The code for the interfaces was developed in TypeScript with Vite and React [88]. It is attached in the Auxiliary Materials Submission Point.
- The hosting is provided by Vercel[79].
- Logging functionality of the mock interfaces is provided with a PostHog API [65].
- Tools used to analyse and collate the gathered data were Python scripts, with the pandas, seaborn, NumPy and Matplotlib libraries [60] [81] [32] [33].

Chapter 1

Introduction

In recent years, more and more of the daily interactions of the average person take place through a digital interface. Many businesses and organisations rely entirely on users exchanging information through such systems, leading to an increase in the demand to understand the complex relationship between user and interface. Indeed, the User Experience (UX) sector is expected to experience a growth rate of 17% in the next 5 years [52]. In 2024, Google invested 1 million USD in U.S. Digital Response to develop AI-powered language-access tools [78], highlighting how organizations are increasingly prioritizing accessibility, where strong UX is central to making these systems effective and inclusive.

Fuchs-Frohnhofen et al. (1996) highlighted the necessity of alignment between user and system, stating that *"an important aspect of human-centered system design is cognitive compatibility, which means that the structure of the human-machine interface of the computer should match the cognitive styles of the users."* [27] It has been suggested that understanding the psychology of users is as important as understanding the system's technical design in determining what makes a system successful. The congruence between the idiosyncratic traits of individual users and the visual design of a system has been studied from many approaches. Such models for this interaction started accounting for users' more innate traits in the 1990s [51], which marks the beginning of impactful research in this area. The traits investigated include self-efficacy [1], general personality factors [5], and cognitive style [9].

Modern studies agree that personality, defined by Maddi (1989) as "a stable set of characteristics and tendencies that determine people's commonalities and differences in thoughts, feelings, and actions," [46] has a great effect on how individuals work with an interface. In particular, it has been shown that UI affects users with specific traits differently where, for instance, users with high emotional stability are more influenced by anchoring bias in design than users with higher Neurotic indicators [56]. Alongside this, specific features have been analysed, such as text sizes, buttons [3] or colours [4] to determine which variations specific traits are more harmonious with. Indeed, an individual's cognitive style, which forms a key component of their personality, has also been shown to contribute significantly. Research suggests that engagement styles across various task-oriented systems vary depending on the participants' profiled type [44]. Participants with the "Thinking" type were seen to use posted reviews when deciding on what CDs to buy, whereas "Feeling" types relied more on their personal impressions from listening to it.

Indeed, it has been shown that the user's degree of satisfaction after an interaction heavily relies on the cohesion between their personality and the system design [48]. This highlights the crucial need for a deeper study of this subject, to better inform design decisions in the future, as conversely, not accounting for it can lead to a poor quality interaction. A 2025 study[72] shows that counter-intuitively, increasing the friction in a user interface (UI) might make for a more memorable experience for the user, thereby crafting a better interaction. This further underlines the importance of devoting resources to such research during any system development process. Any developer should primarily ask: "Given the kinds of people we hope to target, what does this mean for our system design?" For instance, Netflix's highly personalised interface adjusts layout priorities and content thumbnails based on individual users' unique behaviour [76]. 80% of hours streamed come from personalised recommendations, demonstrating the significant value that tailored UI can bring.

The downsides of such interplay, however, also have to be considered. Cross sectional studies [89] have shown that students with specific Big Five Personality measures are more likely to suffer adverse effects of using social media. This emphasizes the importance of not only figuring out the target demographics of a system, but also undertaking deeper personality research of this audience. Such processes can benefit organisations in many ways, including reducing costs for training for internal systems, better customer

satisfaction and less undesirable side effects. This is especially important for vulnerable populations such as children, who often suffer harsh long term consequences of using systems that are not aligned to their mental model, and indeed exploit its vulnerabilities.

Combined with the relative lack of prior research in this area, this is one of the reasons a specific focus on social media was chosen. It has been demonstrated that daily use of social media has many detrimental effects including stress, poor sleep and loneliness [34] [36]. It is imperative that we as a society closely understand the long term effects of platforms that 69.9% of people use worldwide [92]. By having an intimate understanding of the interactions between user archetypes and a system, designs can be adjusted to contribute to more positive outcomes. It is hoped that this research can contribute to a deeper understanding of this.

1.0.1 Aims

- To explore existing research and perspectives on user interface design in social media systems and in general, with particular focus on how interface correlate with influence user behaviour, perception, and engagement.
- To assess how users evaluate usability and their own perception of varying social media interface styles through interactive tasks.
- To investigate the relationship between individual user traits and engagement with specific interface designs.
- To critically evaluate the significance and limitations of the collected data and research methods, and discuss implications for future social media system design and further research in human-centred interface development.

Chapter 2

Background

2.1 Social Media

Social media has been catapulted to relevancy in the 21st century. The average person spends over 2 hours there daily [91], with 70% of consumers discovering new products on social media [92], and over 86% [23] of large organisations using social media for engaging with their customers. This indicates a massive shift in the paradigm of societal cooperation and communication, arguably, on a similar scale to the invention of the printing press. Social media has allowed for easier, decentralised spreading of information which has often undermined traditional broadcasters such as TV, radio or newspapers [26]. Any netizen is given a platform to directly express themselves, making finding communities that align with their own niches easier. On the contrary, it has been widely documented that introduction of such "bubbles" of perception often leads to easier spread of misinformation, polarisation and radicalisation [15]. Due to this, social media and its ability to group alike people has been a key factor in many geopolitically significant events. Examples include the recent Nepal protesters' use of Discord [61] or the role of fake news in the 2016 U.S. election [2]. This parallels the role of the printing press, having increased ordinary peoples' awareness of political discourse and challenging the position of the contemporary establishment [19].

Another analogous historical event that may help inform our understanding of the long term impact of social media could be the introduction of money during the early development of human societies. [25] Evidence suggests that universally recognised valuables emerged mainly in long-distance trade networks, where exchange between strangers required a more standardised means. As with social media, the introduction of a universal medium shifted trust from interpersonal relationships onto a standardized system that enables for interactions to take place at greater scale. This is evident today, as social media has been shown to negatively impact face-to-face interactions, with users increasingly spending time on global online platforms instead [38]. The emphasis is placed on the platform itself, as opposed to personal trust and social bonds. The introduction of money has also been suggested to transform societal relationships, [74] allowing those who control it to establish social hierarchies and gain more power. This is also apparent today, where companies such as Meta, Alphabet Inc, or ByteDance control a vast majority of the world's social media, and therefore the discourse that takes place there.

Such parallels are drawn to highlight the impact of social media is likely so profound that it will require many years of hindsight to fully understand it. As a society, it should be our goal to be mindful of the trends enveloping us and, instead of being subject to entropy, to actively steer towards more positive outcomes. The first step to gain control of such a process is to understand how the average person is affected by it, in order to determine why this happens, so that a range of solutions can be suggested. This research will attempt to contribute to our knowledge of the "why" portion of this formula.

There are already numerous, unique and widely-documented negative effects of social media we are aware of. These include increasing the rates of anxiety and depression among users [35], effects on self-esteem and social comparison [80], reduced attention span and worse work performance [34], addiction and compulsive use [6], cyberbullying [36], and increased loneliness [36]. Alongside these, there are likely consequences we will become more aware of as the current generation, who have been brought up immersed in this ecosystem, grows up.

Such consequences could arguably be framed as functional effects or emergent side effects. Some are aligned with the intended purpose of the platform, whereas others are less desirable, secondary outcomes as a result of design. For instance, the latter include the tendency for social media use to be associated with increased loneliness and isolation. Whereas the original intention of social media was to facilitate

human connection, in some cases the opposite effect takes place. Research has shown that social media does not act as a substitute for all psychological benefits of in-person interaction, [47] thereby creating a potential risk for users who spend a lot of time exposed to it. The human psychology is not entirely accounted for and thus the counter-intuitive outcome takes place. Another example of this would be the aforementioned tendency for social media to cause radicalisation. Social media platforms are designed to help people easily find content and communities that match their specific interests, enabling niche topics and subcultures to grow and thrive. As a result, algorithms explicitly prioritize familiar and relevant content, which can unintentionally limit exposure to differing viewpoints and create focused information bubbles [2]. This is problematic because reduced exposure to diverse perspectives can reinforce existing beliefs and contribute to increased polarization. With an understanding of the secondary emergent negative aspects of social media, policies and regulations can be introduced to combat this.

Alternatively, some effects are entirely intentional by the creators of the platforms, by virtue of the mode of operation of social media. The attention economy has become a major part of the market, being valued at over £21 billion in the UK, with the global digital advertising spending estimated to reach £690 billion by 2027 [30]. Attention is vital for platforms to not only raise revenue via advertising but also promote data collection, brand knowledge and brand loyalty. As a result, companies have many incentives to exploit their users' psychology, in order to leverage higher screen times and maximise interaction rates. Social media employs many "dark patterns", including infinite scrolling [72], social pressure tactics, forced continuity, gamification or fear of missing out (FOMO) [42]. Research has shown that social media platforms create reward feedback loops that tap into the brain's dopamine system through superficial rewards, [45] such as likes or notifications. Similarities can be drawn to mechanisms seen in gambling, which are made to reinforce engagement. These make use of humans' tendency to prioritize short term gains over long term benefits, thereby making them a compelling exploitation of our biology.

As these are intentional choices made by businesses to drive growth, incentives will need to be made for companies to curb their negative effect. In this case, however, the altruistic decision and the business-like decision are not aligned since, for example, limiting the addictive aspects of social media will likely reduce overall screen time, thus impacting profitability. This makes designing appropriate policy less straightforward. Indeed, the inadequacies of the current system have been documented[93] [58]. As a result, the need to understand such harmful interactions and their context is stark. There is already some precedent for this with the EU bodies investigating and lodging complaints against platforms such as X, Facebook or TikTok in regards to such deceptive practices [68] [8] [10].

2.2 Digital Environment Design

Whether positive or negative, such effects are the result of many deliberate choices during the design process. This includes content recommendation algorithms, data analysis software, and, finally, UI and UX. UI and UX are a key element to understanding the interplay between the content creator, platform and user in harnessing their clicks and attention. This has been shown through studies where a large amount of users (primarily female participants) perceived the quality of a product to be significantly worse when visual guidance elements were removed from interfaces [87]. This is tied to the well documented halo effect [54], where people often have a bias towards things they find visually appealing. UX is often designed extremely intentionally, in order to give a specific impression of a system, allow more accessibility or increase usability. It has been found that designs fundamentally change user behaviours and attitudes to a system, affecting their cognitive behaviour and performance. The user's perceived usefulness of the system and their willingness to adopt are also in large part influenced by the user interface. In a 2018 study [73], two distinct personality profiles of participants were found (Neuroticism and extra-Conscientiousness), which were then used to design two interfaces where each uniquely catered to an assigned profile. The participants' eye movement when using the interfaces indicated that the experience was significantly better when their interface was aligned with their individual psychological traits. Another study [64] compared the effect of users using a system they can tailor to their own needs, compared to an off-the-shelf commercially available one. It was found that the participants were able to better navigate through the menus and learn how to use the more adaptable system. A practical example is the aforementioned Netflix case. Netflix's interface is highly personalised, where specific elements such as thumbnails and the portion of content playing when users hover on them differ user by user [76]. For instance a user who is found to enjoy specifically romance movies, might be recommended other movies through the framing of the content via a romance lens. This is done by selecting promotional material that suits what they are likely to enjoy about a movie, thereby making them more likely to watch.

Indeed, 80% of hours streamed come from personalised recommendations, demonstrating the power of such a system.

An element that is occasionally highlighted in such research is the role the personality of the user plays in the process. Though it has been disputed whether personality can be entirely used to predict a person's behaviour, the Interactionist approach underlines the importance of both personality factors and situational variables in explaining human behaviour [21] [22]. User interaction with digital systems can then, arguably, be framed through an interactionist perspective [64]. In this case, behaviour is a function of both dispositional traits and situational factors: $B = f(P, E)$. It has to be remembered that personality is something that differentiates all humans very significantly, being a very strong deciding factor in many of the opinions we have and decisions we make in our day to day. [67]. As a result, many theories have been applied on the challenge of assigning individuals different personality types [85]. This is done by applying different metrics, according to various theories and models [16]. Trait theory is of special significance since it splits personality into several dimensions, which is especially useful for Human-Computer Interaction (HCI) related analysis. It allows researches to isolate specific aspects of participants' psyche efficiently. It has its origins with Eysenck (1963), which specifies how personality can be split into three dimensions: Psychoticism, Extraversion and Neuroticism [24]. Further studies elaborated on this, producing a model with sixteen different personality factors (16PF) [12]. However, continued factor-analytic research suggested that these numerous traits could be further reduced into five broad and robust dimensions, forming what is now known as the Big Five personality traits. This led to the general acceptance of the Five Factor Model (FFM) [17], [50]. The FFM defines personality using the following variables: Openness, Conscientiousness, Extraversion, Agreeableness and Neuroticism.

Outside of trait theory, other models include the Learning Style Inventory [37] and the Myers–Briggs Type Indicator [53] [57].

The Life Styles Inventory is grounded in a social–cognitive framework of personality and behaviour. It assesses individuals' thinking and behavioural styles by grouping them into constructive, passive/defensive, and aggressive/defensive orientations. Rather than focusing on personality preferences, the LSI emphasizes how people typically think and act in work and interpersonal contexts. This makes it consequently less useful in computer interaction research. The Myers-Briggs Type Indicator (MBTI) is based on Carl Jung's theory of cognition, which categorizes the participants on the distinct scales of Extraversion/Introversion, Sensing/Intuition, Thinking/Feeling and Judging/Perceiving.

This research will be restricted to trait-based theories, due to their test-retest reliability and consistency compared to alternative approaches. [49] [63]. A significant criticism of the MBTI is its reliance on binary dichotomies for personality types. It arguably oversimplifies this measure into rigid categories, thus accurate and reproducible user modelling difficult. In HCI research, the Big Five model is widely used and recognized as a more comprehensive model of personality across years of study, as it measures each trait on its own scale, instead of assigning an arbitrary split [56] [51] [90].

It can be said that personality traits reflect variations in how individuals regulate their behaviour, pursue goals, and respond to feedback [18]. For example, individuals with high Extraversion are motivated to chase goals because they are more sensitive to the potential rewards and excitement of succeeding. Conversely, Conscientiousness is more correlated with focus on long-term goals, as external motivations do not have as big of an effect. Utilizing this personality model, an analysis can be conducted to explore the relationships between a user's traits and their preferred design philosophies.

The emotional state of a user further biases and acts as a proxy for exploring their perception of a system. Feeling frustration can indicate a flaw with the system while a positive mood shows the user was satisfied with the interaction. As aforementioned, different interface elements such as colours, layouts, fonts and so on provoke different immediate emotional responses based on the individual parameters of the users. These then dictate how usable each user finds the system. Therefore in order to measure such a perceived emotional state, the Positive Negative Affect Schedule (PANAS) psychometric scale was selected. This is because it captures the user's current fleeting emotional state, alongside the more long term Big Five traits. PANAS is a highly validated and reliable scale [82] that further separates and measures both positive affect (e.g. excited, alert) and negative affect (e.g. upset, jittery). It has a precedent in usage for usability studies [39] [62], allowing researchers to gather empirical data on how a given interface makes participants feel.

Beyond each user's subjective impression, standardized usability surveys provide quantifiable, benchmarkable data. This is significant as it can be used to objectively compare the performance of different UI styles across diverse user profiles. It allows the user to assign the interface attributes, without judgement on the effect of those attributes. The UEQ (User Experience Questionnaire) is the ideal tool for this scenario as it measures such attributes, including both pragmatic qualities (dependability, efficiency,

perspicuity) and hedonic qualities (novelty, stimulation).

pragmatic quality (usability aspects like efficiency and ease of use) and hedonic quality (emotional aspects like stimulation and pleasure).

It is widely accepted in usability research, having been employed for capturing not just if an interface platform works, but whether it is attractive and engaging to the specific participant [39] [7].

While there is a large body of research on the usability of e-commerce sites, productivity/learning software and general web design, the specific UI interactions of social media platforms are somewhat under-examined in these usability contexts.

Although investigations have been conducted on the relationship between the user and specific elements of social media interfaces, there is less research mapping general social media UI patterns to individual personality traits and emotional responses, despite these platforms forming the primary digital environment for many in the modern world.

Chapter 3

Methodology

3.1 General Approach

For this section, the previously introduced concepts and ideas will be used to describe how this study went about fulfilling the set objectives:

- To explore existing research and perspectives on user interface design in social media systems and in general, with particular focus on how interface correlate with influence user behaviour, perception, and engagement.
- To assess how users evaluate usability and their own perception of varying social media interface styles through interactive tasks.
- To investigate the relationship between individual user traits and engagement with specific interface designs.
- To critically evaluate the significance and limitations of the collected data and research methods, and discuss implications for future social media system design and further research in human-centred interface development.

To gather data and assess this, the most straightforward approach was to create a mock user interface and have users interact. Each user would take a questionnaire beforehand, to classify their personality traits into pre determined categories, alongside some usability questions. This is a very common and tested approach [3], [73], [75]. However, there are many variations on such an approach. Firstly, the mock interface needs to vary in very specific dimensions only, to make sure that the effect of changing certain variables produces the alleged effect for a given personal trait. A basic format of this is utilizing a single UI, with only a single variation that is shown to participants of different determined categories, otherwise the systems would be identical [41]. This allows for focus on investigating the effect of very specific UI elements. However, since it would be necessary to narrow down a small number of different elements to investigate, this somewhat limits the scope and extent of the investigation. Therefore the approach of creating three different systems that target a narrowed selection of categories of Big Five traits was chosen. There is precedent for this [7], [56], [73], as studies have created 2 different UIs across scales such as Agreeableness, Openness to Experience, Extraversion/Introversion and so on. Due to the limited potential number of participants, the number of mock interfaces had to be chosen carefully. This is to optimise the scope and range of investigation, while ensuring that a sufficient number of participants of each trait interact with and see a given interface. When designing this study, the number of participants was not anticipated to be more than 50, therefore the number for the unique interfaces was chosen to be 3. This is in congruity with the 3 different personality trait groups. An interaction matrix is generated from this, as seen in Table 3.1, meaning the most optimal sample of participants and allocation would result in no more than 33 participants for each cell as each participant is given 2 interfaces. This is a reasonable amount for a study of this scale in order to get a sample of sufficient size to be able to make justified conclusions on the pattern present. It would also give a sufficient sample of the scenario where the user's personality focus aligns with the focus which their UI caters to.

As aforementioned, the Big Five personalities, that being Openness, Conscientiousness, Extraversion, Agreeableness and Neuroticism, were decided to be used as indicators of the participants' personalities. The Five-Factor Model (FFM) is a widely accepted model of personality, as it offers a standardized

Personality Cluster / UI Type	**High Extraversion, Low Neuroticism**	**High Conscientiousness, Agreeableness**	**High Neuroticism, Low Extraversion**
UI C – Expressive, Hot	High preference; enjoys fast pace, visuals, instant replies, low friction	May feel chaotic or distracting; lacks structure and control	Potentially overwhelming due to high sensory load and fast interactions
UI B – Structured	Moderate engagement; may feel slightly constrained but still usable	High preference; aligns with need for order, clarity, and rules	Comfortable; predictable system reduces stress and uncertainty
UI A – Open & Anonymous	Lower engagement; lack of identity and friction reduces social reward	Mixed response; high control but friction (CAPTCHA, rules) may frustrate	High preference; may feel attracted to less explicit social norms and increased anonymity

Table 3.1: Predicted interaction preferences between UI types and personality traits

framework that integrates a large portion of research on individual differences. Determining the number of the personality clusters and how to group them was therefore an important decision. It was decided that the number of such clusters would have to be 3, in order to accommodate for the scope of this study. A selection of the Big Five Traits would therefore be used for each personality cluster. This is to make analysis more convenient as it would allow for each user's personality result to determine whether their personality aligns with the UI's they interacted with.

Moreover, the kinds of tasks selected for the users to perform were based on existing research. The users were made to perform tasks that they would usually do on the types of interfaces being tested. For instance, a study was performed on the efficiency of using personality aligned layouts for a studying app [4]. The experimental tasks asked each participant to practice for tests and take quizzes to improve their grades. Since this is what a user expects to do in such a system, it allows them to immerse themselves in the presented scenario and therefore give more faithful answers. In parallel, another study [75] analysed the efficiency of using a personality - aligned Nuclear Power Plant operating system. Participants' tasks were made to reflect the typical duties of a plant operator, such as monitoring and detection, situation assessment, response planning, and response implementation. As a result, the tasks for this study were created to reflect what a user would do in each specific system. More detailed breakdown is given in each interface's section, but typical tasks administered include searching for a specific user, sending a general reply to a message, scrolling the feed and replying to a specific post.

A significant decision to take was also the allocation of the different user interfaces for each participant. Due to limitations for the time for which a single study can run, as well as potential participant retainment issues, there are more interfaces than a participant can see per session. It was therefore necessary to create some kind of assignment rule. Prior studies often administer personality questionnaires in advance of the users seeing the interfaces, in order to show each participant the interface that either corresponds to the participant's profile or is in conflict with it. [4], [89] These studies often only show each user a single interface and only examine the effect of changing the UI across a single spectrum (Introverted / Extroverted, linear / non linear layout). In contrast, this study's multipolar approach makes this method less useful since there are more variables at play. The aim is to assess the reactions of a range of personalities to a range of interfaces therefore it is advantageous to spread each participant's workload over more than one interface. As a result, the number of interfaces a given participant would interact with was settled on 2. The assignment of these interfaces was another consideration.

The first option is giving each user the Big Five questionnaire first, calculating their score on each of the scales and only then assigning them an interface. This would likely be in the form of a dynamic algorithm that would distribute the participant - interface pairing according to the number of responses received so far for each combination. However, this would rely heavily on the algorithm having accurate and up to date information on the amount of responses received. As this questionnaire would have a relatively small number of participants, this would cause challenges since the algorithm might not be able to account for participants who are completing the questionnaire at the same time. As a result, the assignments might be skewed which would impact the quality of the dataset since the relatively low participant number makes evenly distributing them even more significant. Moreover, an element of confirmation bias would likely be introduced with this approach, since it would be assumed that the hypothesized personality clusters actually correspond to the UIs designed. It might be the case that, for instance, UI B is in fact preferred by individuals that are high in Neuroticism.

The second and simpler option was not taking into account each user's personality question responses at all. Instead, the interface would be assigned by a random trial, with 1/3 probability of receiving each for a given participant. The only constraint on this would be that a participant could not receive the same

interface twice. This would eliminate any threat of making assumptions, since the random trials would distribute it according to the probabilities, therefore trending towards each interface being shown an equal amount of times. Moreover, it also makes the questionnaire process simpler as no extra clustering and assignment calculations need to be done, thereby removing a vector for biases and errors to enter. A limitation both methods encounter is that it is difficult to guarantee an even split of each personality cluster in the surveyed population. This could be partially remedied by sampling diverse demographics, as it has been shown that certain categories of the sampled demographics, such as gender and age, tend to be predisposed to certain Big Five traits. [83].

Overall, the second option was chosen to tackle the issue of assigning UIs as it is less likely to poison the well of the results, as well as its ease of implementation.

HCI Aspects	**UI C – Expressive, HOT**	**UI B – Structured**	**UI A – Open & Anonymous**
Primary Interaction Focus	Camera / Messaging	Messages / Content	Content
Transitions & Animations	Fast animations, click sounds, animated loaders, strong hover effects	Moderate animations, no sounds, subtle hover effects	Minimal or no animations, no sounds, no hover effects
Disappearing Messages	Enabled by default, visible expiration cues	Disabled, all content stays	Threads expire without timers (time posted shown)
Message Positioning	On one side	Left/right aligned by sender with threads	Chronological vertical alignment
Reply Behavior	Instant replies and typing indicators	Delayed replies	Need to verify to reply
Error & Recovery Communication	Large, visible undo buttons, bright and reassuring tone	Smaller undo button, neutral informational tone	Minimal or no undo; cold, system-style tone
Error Button Design	Visible X or undo, high contrast	Standard-size buttons, neutral colors	Small or hidden error controls
Norm Visibility & Social Rules	Minimal explicit rules, moderators not visible	Community rules clearly visible, moderators present	Less visible rules, moderators present
Anonymity & Identity	Faces on profile image	Real name and identifiable profile	Anonymous
Conversation Rendering	By most liked	Mostly linear with mild reply structure	Chronological
Friction to Interact	None	Rules visible before posting	CAPTCHA, rule acknowledgement per post
Perceived Social Risk	Low	Medium	High
Sensory Load	High	Medium	Low
User Agency & Control	Low–Medium	Medium	High

Table 3.2: Comparison of three UI design conditions across HCI aspects.

As described in Table 3.2, each interface targets a specific niche in the personality space. The Big Five personalities targeted were hypothesized at this point, and so a partial role of the experiment would be to confirm this. Something that was discovered along the way was that conclusions or ideas made for elements of design that encourage certain behaviours or cater to a certain demographic already correspond to what has been implemented, in a kind of convergent evolution. For instance, the concept of an extroverted-focused interface that opens at a camera screen was used because research shows that extroverted people enjoy looking at human faces more than others [71]. Indeed, this is something already implemented by platforms such as Snapchat. In addition to this, AI was used in a limited fashion during the design process. The Gemini LLM was consulted about the kinds of topics, words and phrases that one might encounter on the social media inspiration was drawn from. For instance, a prompt would be asked such as "Give me some examples of typical post titles and comments on Instagram" or "What kinds of topics would the posts of a given user's feed page likely talk about". This was done in order to minimize

the personal biases of the author, when crafting the content in the interface. Creating content that is similar to what the user likely sees on the real-life system is more likely to make them feel immersed in the tasks and thus give accurate answers. AI is especially useful for this tasks since it likely has been trained on the contents of these websites and the LLM is thus likely very accurate in giving a general idea of what a typical Instagram feed looks like. Each interface will now be described to justify its design, drawing on previous research.

3.2 Interface A

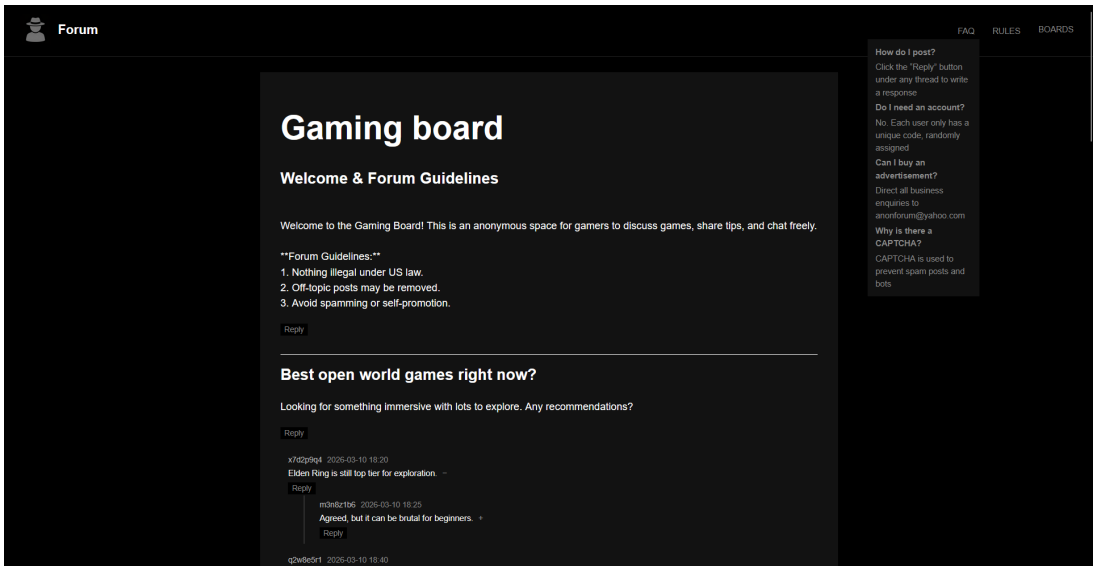


Figure 3.1: Feed page of the Gaming Board on Interface A

The main real-world inspirations are Reddit, 4chan or specific subject boards such as Stack Overflow.

Interface A was the simplest to design. This was since its simplicity is the factor which appeals to the more introverted, preferring clear, structured, and low-stimulation interfaces [4]. The main colour on display is black, as this has been shown to influence a high Neuroticism trait [43]. It features sharp container corners and sparse HTML formatting. There is very little personalisation options, such as an account of private messaging. The whole board is entirely anonymous and open, meaning each user can see exactly the same main page. This is a vital aspect of such account-free pages where a sense of identity is more volatile. Each board is likely to be shared by more disparate users (unlike in a platform similar to interface C, where the feed algorithm would ensure each "interest bubble" would be more likely to be only inhabited by people of similar dispositions). This aligns with the Openness trait, as a typical user of this interface is likely more comfortable encountering polarized opinions and diverse perspectives.

Another element of this interface is the de-personalised aspect. As this was designed to cater to introverts, the presence of other humans interacting with the same content is de-emphasized. This is done via removing elements such as profile pictures or usernames, thereby dehumanising the other "users". Moreover, this site does not include any indicators of any likes or comment numbers. Often, the number of comments and how many people are viewing a certain post creates a sense of anxiety about posting, which discourages introverts from posting [86]. All posts are displayed in a chronological order, giving the impression of an open platform with no favouritism or social hierarchy. The content itself is emphasized, and the social aspect of interaction is de-emphasized.

All comments are further shown in a highly structured way via threads. A user can reply to any comment directly, adding to this web of conversation. As no metrics are shown besides each comment, each reply has no "community approval metric". Typically very controversial statements often attract a high number of replies, which is another way such platforms incentivize polarisation and extreme statements since it is an obvious way of attracting attention. This is an aspect of this interface that plays into a Neurotic trait [59].

The content within the system was also considered. As the personality targeted was the Neurotic, Open and Introverted, the kinds of hobbies preferred by this population were included, as to reflect a

real forum. Indeed, these topics correlate to the content in forums this interface was inspired by. This includes topics focused more on media, technology and hobbies as opposed to people and relationships. Examples include video games, movies, programming etc.

The Openness of the board is further emphasized on by the loose rules that are only displayed briefly as the user scrolls (unlike in interface B where they are on display at all times). This creates a loose attitude in terms of the social rules and expectations present, resulting in exacerbating the Online Disinhibition effect [56]. The anonymity allows the user to be less accountable for what they say, while the online environment makes them feel "invisible" to the judgement of others in contrast to a physical conversation. As a result, the users feels safer and thus more likely to say riskier statements, as correlates with their Neurotic type.

A further design philosophy is the high friction to interact with the system. Similar to other such social media systems, each user has to first complete a CAPTCHA in order to post. While in these systems the purpose of this is to prevent bots and spammers from targetting the forums, it introduces an interesting variable in the way a user interacts with it. It presents a relatively high cognitive barrier to entry, which might discourage individuals with less Openness to interact. It makes replying to posts more of a task, therefore encouraging longer (though not necessarily more thoughtful) replies.

3.3 Interface B

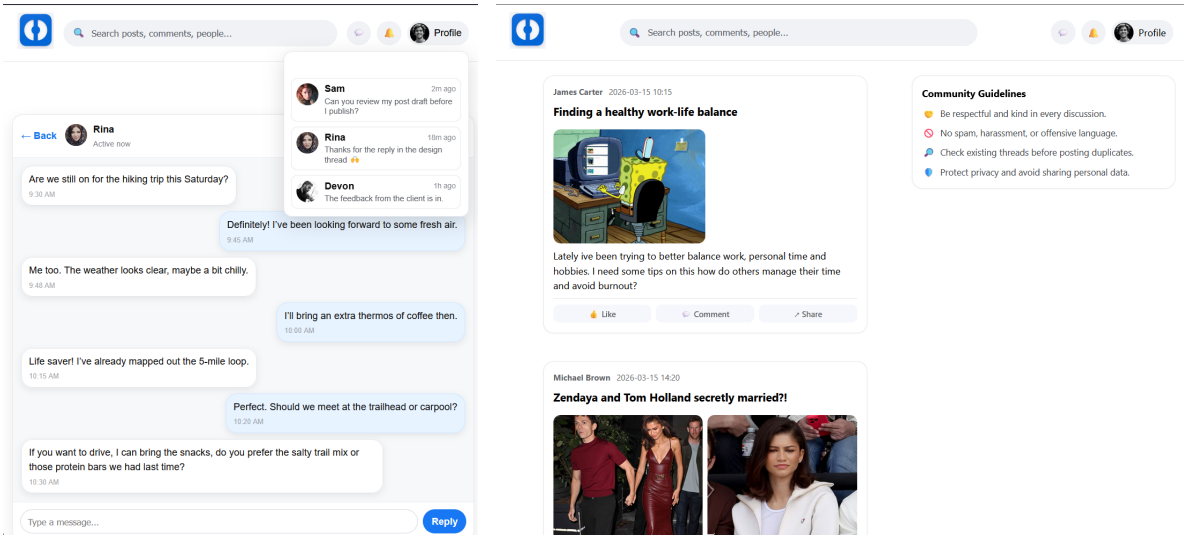


Figure 3.2: The messaging interface and main feed of interface B.

The inspirations for this interface were LinkedIn, Facebook. It overall presents a clean, friendly and organised look that presents agreeable content and targets users who exhibit less Neuroticism than interface A.

The colour scheme for this interface was chosen to be blue. As supported by studies [20], the colour blue encourages calm and collected thoughts, which is something the system wants to reflect. Moving from interface A, it features more rounded corners with subtler container transitions utilising shadows and gradients, thereby reflecting the calmer and more structured environment. This is something that appeals to individuals high in Conscientiousness. The organised sense is further emphasized by the presence of a search bar, that lets the user specifically find what kind of post or user they are looking for instead of having to manually scroll to search.

There is an increased sense of personalisation in this interface. The user has their own account page, as well as a simple account profile photo. This serves to appeal to individuals with a lower Neuroticism trait, since an increased sense of social responsibility is present. Although most posts do now show the profile photo of the poster, clicking their profile allows the user to see a semi-identifying profile photo. This was to place this on a scale between a system fully tied to one's real persona (interface C) and one fully divorced from it (interface A). It introduces a sense of social responsibility as now, any person can see some details about who made a post or a comment. This discourages polarising content and encourages agreeable posts and replies, appealing to the intended demographic.

This is also reflected in the rules present, since now, the rules do not disappear as the user scrolls. They stay to the left of the feed, such that they are always visible thus making them more apparent to the user as they are interacting. Though they are not very different from rules given for interface A, their omnipresence is likely to influence each user in the kind of content they decide to publish.

The friction to interact is also decreased, as indicated by the lack of a CAPTCHA, thus encouraging users to post more, leading to shorter replies under posts and more agreeable content. The purely chronological structure of interface A is also removed. Now, posts are shown in a order dictated by what the user can assume to be an algorithm, as they are logged into an account.

The kinds of posts the user sees also fit into the Conscientiousness paradigm, reflecting more the type of content one might expect on a platform such as Facebook or LinkedIn. The feed emphasizes socially relevant, broadly appealing content, such as updates from friends or colleagues, professional advice, local news, pop culture trends, event announcements, and lifestyle tips. For instance, a user might see posts about community volunteering opportunities, career development articles, popular music releases or local restaurant reviews.

This contrasts with interface A that caters more to niche interests or specialized communities, where content is often specific and topic-focused (e.g. detailed technical discussions or hobby discussions). In comparison, this interface avoids overly narrow content, instead promoting posts that foster general social cohesion and shared experiences.

Another way in which the user's session is made more personal is by introducing direct messages. In this system, the user has a variety of contacts to choose from to message, further enhancing the personalisation aspect. Closer interaction with other users who are more explicitly shown to be human with real names and semi-anonymous icons further makes this interface more suitable for an extrovert audience. The layout of each message sent is also safe and predictable, utilising the standard left - right layout seen in the majority of platforms such as WhatsApp, Messenger etc.

3.4 Interface C

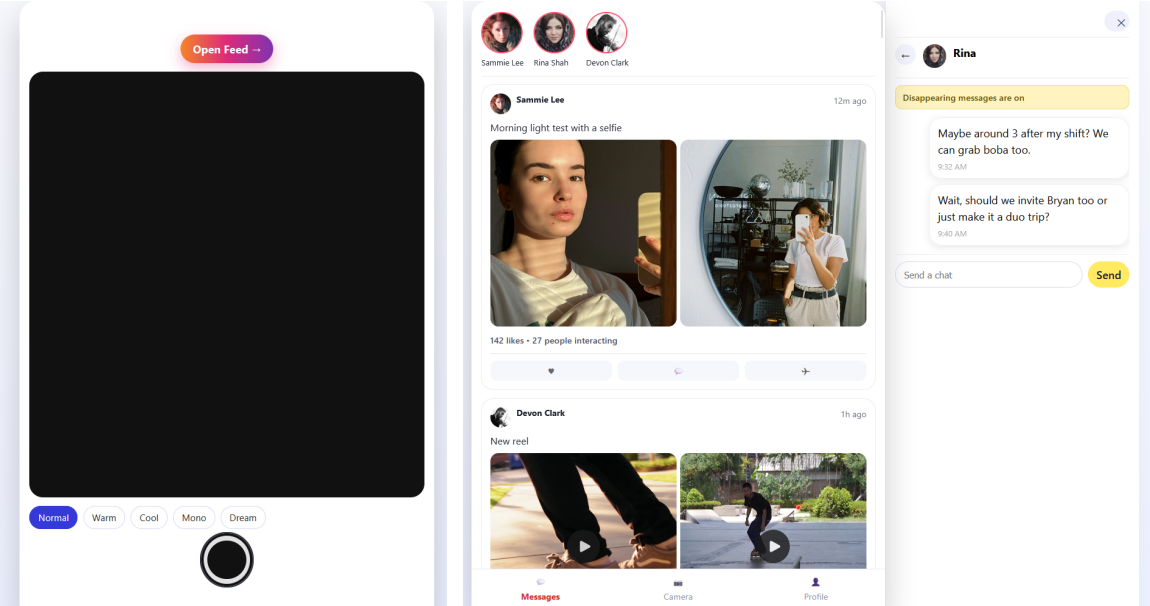


Figure 3.3: The front-page (left) and main feed (right) of interface C, scaled to equal height.

The primary inspirations for this interface were Instagram, Snapchat and similar camera based chatting interfaces such as Locket or BeReal.

The colour scheme of this interface is primarily a gradient featuring purple, red and yellow. All of these are very hot and vibrant colours that typically appeal to extroverted individuals [20]. Indeed, it is immediately visible that this interfaces takes a slightly different approach, as the main landing page is not a feed but a camera. Extroverted individuals enjoy looking at photos of humans whether it be themselves or others [71], therefore this is another way to cater to this demographic. The user can scroll via various filters, which then allow the user to take a photo of themselves and "post" it to their profile.

The main feed is similar in many ways to the more extroverted driven interface B, though it features more visually attractive elements. For instance when pressed the buttons emit a coloured gradient and vibrant animations are played as the user interacts. This is an extrovert driven design perspective with such explicitly dynamic elements. Another push towards the extreme of this axis is the posts, since each now features a fully identifiable profile photo with a face that is present in each post. At this point anonymity has essentially been eliminated in favour of a bigger sense of social responsibility. As each user's real identity is fully and always visible to others, less extreme and polarising statements would be made. The typical comment and post length also decreases since there is a lower barrier to entry. Virtually no friction to interact is present thereby encouraging all members to interact as much as possible. The visibility of the number of likes on posts as well as the sorting of posts by likes further makes a hierarchy clear, since posts with more comments and likes have more influence. A sense of social need of approval is brought about especially explicitly in this interface, appealing to the users' low introversion [70].

Another element to mention is the messages. The messages do not appear over the main feed, but instead exist side to side with it. Again, this encourages a connected feel, sacrificing a sense of compartmentalisation. As a result, the user would be someone lower in Conscientiousness or preference for compartmentalization, as the interface prioritizes social connectedness over strict organization. [48], [73]

In a subversive fashion, the messages are all positioned on one side of the panel. Moreover, the messages disappear after they are read. This design favours users with lower attention span and higher Extraversion, as it encourages rapid scanning of content rather than continued, focused interaction with each message. Again, this goes against the trait of Agreeability. Since past conversations cannot be seen as one scrolls up, spontaneity is encouraged with replies. This could be seen as a more accurate reflection of a real-life conversation, thus being a more attractive feature for users high in extroversion.

The information density in this interface is also relatively low, in line with findings [4] on the correlation with Extraversion.

3.5 Questionnaire

The primary question block concerns the users age, gender, as well as how many hours they spend on social media a week. This helps interpret the results, as users who spend fewer hours on social media may be less familiar with the dynamics and conventions of different platforms.

In order to achieve objective 2 and 3, it is necessary to gather three types of data. Primarily, the personality of the user needs to be identified and categorized, so that it can be assessed against any preferences. As mentioned, this is achieved using the Big Five trait survey. Specifically the Ten Item Personality Measure (TIPI) [31], as it is designed to be a quick series of questions to fill out. This is to keep the study length appropriate so to minimize participant fatigue, while maintaining high psychometric convergence with more extensive Big Five instruments.

Nextly, the user's perceived usability of each of the systems needs to be surveyed. It is vital to understand what attributes the user would assign to the system. For this, the User Experience Questionnaire (UEQ) [40] will be used. The users' impressions of the interfaces will be gathered to split them across traits perceived by each personality cluster.

However, the subjective element of this study is further underlined with the need to account for the fact different people will likely find different aspects of a design appealing. For instance, someone might find a predictable interface appealing more than an unpredictable. Likewise, a boring interface might just be what a user is looking for, which is not accounted for in the survey so far. For this reason, it is necessary to introduce the PANAS emotional response [82]. This is to make sure that not only each user's assessment of the system is present, but also which features they personally identify with.

3.6 Participants

The total number of participants was 40. Participants were recruited primarily through online survey platforms such as Prolific, as well as through more personal methods, such as word-of-mouth or direct invitations. Though there is potential to reach out to more participants personally, there is a chance this will be a biased sample. Consequently, Prolific was used to gather additional high quality responses to even out the potentially skewed data. Each Prolific participant was compensated £3 for their time taking the study. Median time to finish the study was 19 minutes.

3.7 Apparatus

The survey hosting website used was Qualtrics due to its strong academic background and usage in a wide range of similar research [66]. The code for the interfaces was developed in TypeScript with Vite and React to prioritise a lightweight architecture as significant backend is not needed. The hosting is provided by Vercel as a simple frontend is all that is required. Moreover, the logging functionality is provided with a built in PostHog API, as this allows for all actions taken by a participant to be measured, including how long a post was viewed, the text sent in messages, the button clicked, the time spent on a given web page etc.

Chapter 4

Result Analysis

4.1 Analytical Tools

The primary tool used to analyse and collate the data was Python, with the pandas, seaborn, NumPy and Matplotlib libraries. Python is a convenient solution to provide scalability as the total size of the logs to be analysed was over 20MB, while pandas was selected for its ability to perform complex data merges and joins. Matplotlib and seaborn further provided facilities for specific control over data visualisation for Exploratory Data Analysis and presentation. The scoring method for each instrument (PANAS, TIPI, UEQ) was implemented as outlined in their relevant documentation [82] [31] [40]. Additionally, AI was used in a limited fashion to generate the code to create and stylize the graphs seen below

4.2 Broad Behavioural Relationships

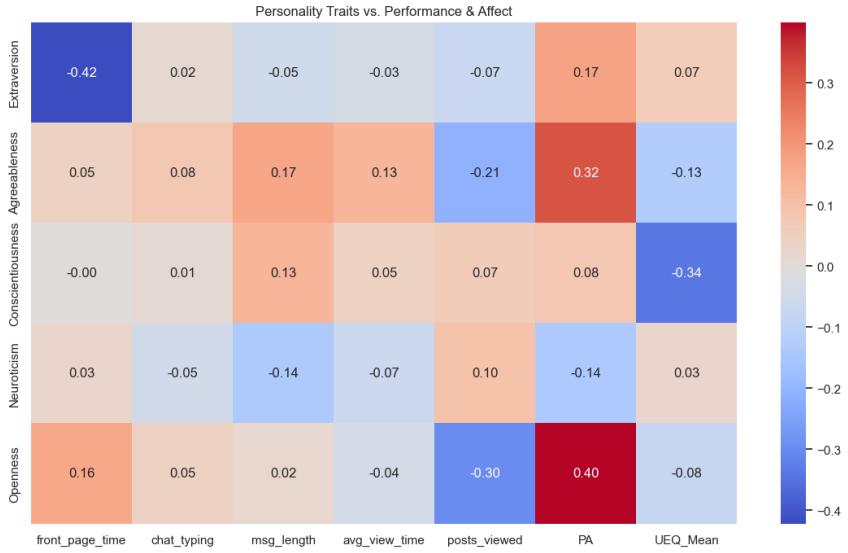


Figure 4.1: Correlation matrix between collated behaviours, scores and each of the Big Five Personalities

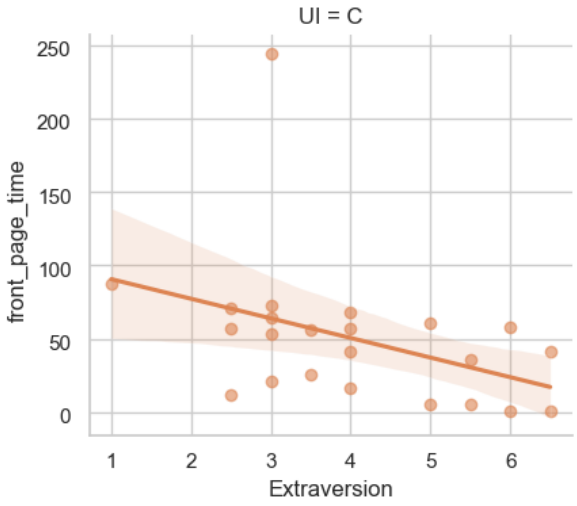


Figure 4.2: Scatter graph between the time a participant spent on the interface C front camera page and their Extraversion score

The primary relationship to discuss is the interplay between the user's personality traits and their interaction with the interfaces. The initial analysis will, therefore, take place along those lines. Figure 4.1 displays a correlation matrix that will allow for analysis of the behaviours of individuals who exhibited a high degree of the given personality.

The heatmap reveals that individuals high in Openness correlate with a Positive Affect (PANAS+ score) (r = 0.4), suggesting that these are the users who are curious and value novel experiences. As a result, they may find any given interface more engaging and therefore are more satisfied. In addition, the Conscientiousness traits shows a notable negative relationship with Usability (as measured by the UEQ Mean Score) (r=-0.34). This indicates that the detail-oriented and organised users were especially critical of the interfaces. With more time, the interfaces could be more polished and more features added such that more Conscientious users are not discouraged by the imperfect mock-up nature of the systems.

The Agreeableness of a user appears to have a moderate correlation to the PA (Positive Affect) score (r=0.32). This underlines how more cooperative and amiable users have a more favourable response to the systems, in contrast to Conscientious users who might struggle more with less specific tasks and vaguer instructions.

A notable negative correlation exists between Extraversion and front_page_time (r=-0.42). As visualized in Figure 4.2, participants with higher Extraversion scores spent less time on the landing page (Figure 3.3), when they were selected to test interface C. As a result, this suggests that highly Extraverted individuals may favour more active interaction as opposed to static tools such as the camera interface. Extraverts generally have a higher threshold for cortical arousal [29] and get bored faster with static or solo activities, which might motivate them to move on faster from the initial screen. This refers to the baseline level of brain activity, which in extraverts is typically lower, requiring them to seek more external stimulation to reach their ideal level of cognitive performance.

This is in contrast to the hypothesized relationship, since studies have found that that Extraverts enjoy looking at people's faces more [84] and thus might linger for longer.

There is also a noteworthy lack of relationships to highlight. Interestingly, Conscientiousness (r=0.01) and Extraversion (r=0.02) showed almost zero correlation with chat_typing speed. This suggests that the action of thinking of a message and typing it out may be driven by other variables, rather than just these specific personality traits.

Neuroticism (r=0.03) and Extraversion (r=-0.03) demonstrated a negligible relationship with average_view_time for each post. This implies that the focus on an individual on a post remains relatively consistent across these personality traits, regardless of the user's emotional stability or tendency to be socially outgoing.

Conscientiousness showed a zero correlation with front_page_time, indicating that a user's level of self-discipline has no bearing on how long they spend on the initial interface screen.

4.3 Person-Environment Fit

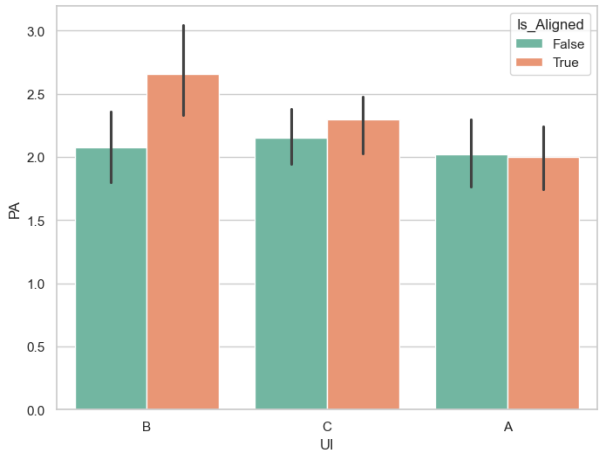


Figure 4.3: Bar chart showing the Positive Affect score for each participant with either an aligned or non aligned personality profile per each interface

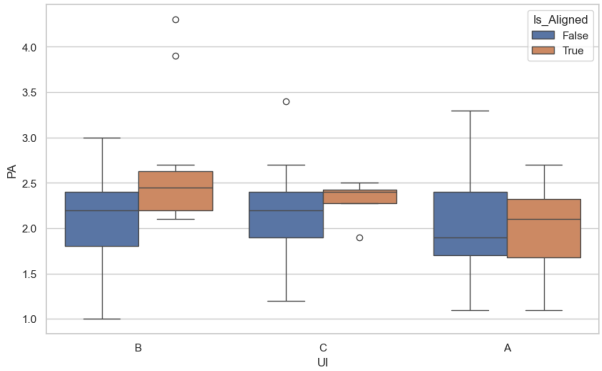


Figure 4.4: Box plot for each bar depicted in Figure 4.3

The bar chart on Figure 4.3 demonstrates a relationship between the "Aligned" users, those whose personality clusters matched the intended design of the UI, and their reported Positive Affect (PA) compared to non-aligned users.

A trend is most pronounced in interface B, where Aligned users reached a mean PA of 2.7, compared to 2.1 for Non-Aligned users. This suggests that for certain interface styles, matching the design to the user's trait profile slightly enhances emotional satisfaction.

Figure 4.4 also reveals that interface B (Aligned) not only has a higher median PA but also contains a number of high-scoring outliers (above 4.0). This indicates that for a subset of users, the "fit" was very effective.

Conversely, interface C shows a much tighter distribution for Aligned users. This implies interface C produces a more reliable or consistent emotional response when users are matched to this specific UI. The median PA score is slightly higher for aligned users than non-aligned, though the relationship is not nearly as pronounced.

Interestingly, interface A further diverges in this. In both Figure 4.3 and Figure 4.4, there is a visible lack of a relationship between alignment and increased Positive Affect.

For UI A, the mean PA for Aligned users (2.0) is nearly identical to or even slightly lower than non-Aligned users. This suggests that the design of interface A may be "personality-neutral" or that the specific traits targeted for this UI did not successfully translate. The presence of other confounding variables might have further caused this digression. These will be discussed in the Critical Evaluation section.

Further analysis shows that the users of interface C in the top 25% of the Positive Affect score also score slightly higher in the Extraversion (4.6) trait than the average (4.2). This indicates that

perhaps the measures to mimic real life social interactions with this interface in fact do attract people more comfortable in such settings. The elements that contribute to this primarily are the reply behaviour, where disappearing messages create a sense of spontaneity about the conversation. The increased presence of posts and profile images with people's faces further goes to reminding the user that they are interacting with real people, instead of other users "hiding" behind avatars.

In addition to the the Figures, t-tests and a two-way ANOVA values were calculated to assess the significance of these results. An independent-samples t-test was conducted to determine if users aligned with their targeted interface experienced higher Positive Affect (PA) than non-aligned users. While aligned users reported a higher mean PA (M=2.32) compared to non-aligned users (M=2.08), this difference did not reach standard statistical significance, t=1.72,p=0.091. However, the result demonstrates a marginal trend in the hypothesized direction, suggesting that personality-interface fit may positively correlate with user satisfaction, though a larger sample size and a study of more variables would be required to confirm this effect.

There was no significant main effect of personality clusters on PA (p=0.892), confirming that baseline affect did not inherently skew across personality groups. Furthermore, the PA of an interaction between an aligned interface variant and personality cluster was not statistically significant, $F(4) = 0.8, p = 0.587$. While there was a marginal trend suggesting that specific UI designs alter user affect ($F(2) = 3.14, p = 0.081$), the data did not reveal a definitive interaction where specific personality clusters uniquely peaked in satisfaction only within their theoretically assigned environments.

The data suggests that Personality-Environment fit is not a universal pattern in this study. It is somewhat effective for specific layouts (like UI B) but may be negligible for others (like UI A). This reinforces the complexity of user experience: a conventionally well crafted UI can sometimes override the need for personality-specific tailoring.

4.4 Design Validity and Real-World Parallels

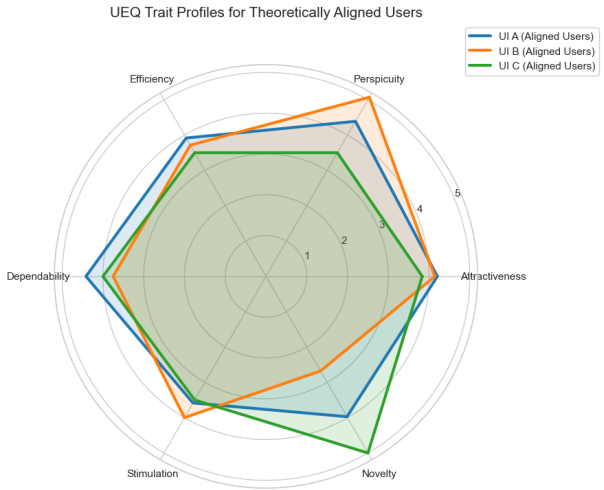


Figure 4.5: Radar chart showing the UEQ trait selection of users in the top 25% Positive Affect score for each interface

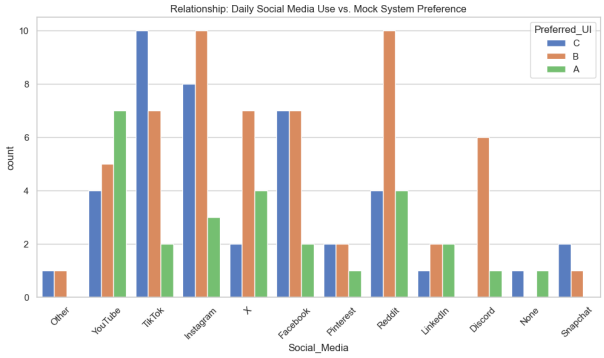


Figure 4.6: Bar chart showing the interface preference for each participants per their real Social Media usage

Figure 4.5 demonstrates that personality-aligned users experienced the three interfaces through distinct usability attributes. As aforementioned, one user might appreciate the Perspicuity scale more than another. In this graph, the alignment for a user and personality was not assigned based on the theoretical clusters. This is because this was not shown to be a strong predictor for alignment in satisfaction. As a result, the UEQ results of participants who had a high Positive Affect score for each interface were plotted. interface A was seen to have a slightly lower perceived Stimulation rating, being primarily as a tool of Efficiency and Dependability. This suggests it successfully functioned as a utility-first environment.

Interface B excelled in Perspicuity (clarity), Attractiveness, and Stimulation. This indicates that these users of UI B found the system not only easy to understand but emotionally engaging and aesthetically pleasing. In contrast to the hypotheses, however, interface B was perceived to be the most stimulating interface.

Interface C also shows a unique niche by scoring highest on Novelty. This confirms that the design succeeded in feeling "hot" or unconventional, which likely appealed to the High Openness traits identified in earlier sections. Alongside this, it was also shown to be the least Attractive and Efficient, though not by a significant value. This could have been due to the limited scope of UI C, as it did not have as many features as the users might have expected of such a system.

Notably, all three interfaces show a lack of variation in the Dependability metric (4.0 on the radar chart). This suggests that regardless of aesthetic or layout differences, users found all three systems reasonably technically stable and reliable, ensuring that usability frustration was not a confounding variable in the study.

As shown in Figure 4.6, there are some striking relationships between participants' mock-system preferences and the social media they use daily. Users who frequent Reddit, Discord and X showed a heavy preference for interface B. Given UI B's high "Perspicuity" scores, this suggests these users prioritize clear, organised and community-centric architecture. This partially validates the choice of the design elements in UI B. On the other hand, UI B users also had a high preference for UI C-tangential interfaces (TikTok, Instagram), suggesting that there is significant overlap in the kind of person who uses both.

Nevertheless, interface C also had a high preference for users of high-stimulation, algorithm-driven platforms like TikTok and Instagram. This aligns well with its high Novelty and Stimulation design intention, suggesting UI C successfully captured the high-energy engagement mode typical of these apps.

For certain platforms like Pinterest and LinkedIn, there was a visible lack of a dominant UI preference. The distribution across A, B, and C is relatively flat, indicating that these platforms' unique engagement modes may not be as easily categorized by the personality-led UI variants used in this study.

In conclusion, the overlap between real-world platform usage and mock system preference provides some evidence for the study's Ecological Validity. It confirms that the mock UIs were not just abstract designs but functioned as somewhat effective proxies for real digital environments. This validates the theory that users prefer digital spaces that provide a Personality-Environment fit, similar to the ones they have already chosen in their daily lives.

4.5 Limitations and Caveat

While Figure 4.1 and Figure 4.2 show some associations (e.g., the negative correlation between Extraversion and front-page dwell time, r=-0.42), it is vital to remember that correlation does not equal causation. It cannot be definitively stated that a user's personality causes them to navigate the UI in a specific way; we can only observe that these variables move in tandem.

Factors not captured in this study such as a participant's level of digital literacy, their current mood prior to the task, or even the device used (mobile vs. desktop) might be the actual drivers behind some of their personality scores and their UI interactions.

As noted in the analysis, the t-test comparison of Aligned vs. Non-Aligned users yielded a p-value of 0.091. While this indicates a marginal trend, it falls short of the traditional $p < 0.05$ threshold for statistical significance.

Moreover, the overlapping error bars in Figure 4.3 and the wide spreads in the box plots (Figure 4.4) suggest that with a larger, more diverse sample, these trends might either solidify into significant findings or disprove the hypotheses, highlighting the danger of over-interpreting results from a limited dataset. It must be remembered that the number of participants is a significant limitation in this study.

Chapter 5

Critical Evaluation

Overall, the study proved to be somewhat successful. A relationship was shown to exist between the personality of a user and the kind of social media they preferred. This was shown to a limited extent, as indicated by the p-value for users interfacing with the various interfaces. With a sample size of 40, the results can also be assigned a reasonable confidence, though a higher sample would have been preferred. With a too small sample, there is an increased danger of assuming a false negative. Consequently, this might have occurred during the analysis, when it might have been concluded that there is no effect simply because the sample was not big enough to see a pattern. Indeed, each UI was only seen about 25 times for each given theorised trait cluster. A larger sample would allow for detection of subtler patterns and be less prone to such biases. In addition, with more participants, the study would be more likely to capture the diversity of the actual population. Despite efforts to do this via Prolific, a smaller sample gives more significance to outliers. Due to the Law of Large Numbers, adding more participants causes the average of the results to get closer to the true average of the whole population [41]. This would make the findings more applicable to the real world.

In addition, some of the confounding variables that were not accounted for might have been each user's preference for using phone-based UIs or PC-based UIs. There is a big split between users who enjoy interfacing with digital systems more on PCs and mobile phones [69]. Some users prefer to use mobile phones for smaller lighter tasks such as a survey, whilst others might prefer to use a PC as their main device. As the survey and interface were only designed for a PC layout, any user who might have preferred a mobile layout would have scored lower on the Positive Affect score, than if they could have used a mobile interface.

The high Dependability scores across all UIs in Figure 4.5 might be an artifact of the controlled experimental environment. In the real world, platform glitches, toxic comments, or intrusive ads would likely disrupt the experience in ways this study could not simulate. This might make the results less reflective of a real interaction

Lastly, this data captures a single moment in time. Personality traits are relatively stable, but User Affect is somewhat volatile. A participant who had a bad day might rate UI A poorly, not because of its design or their personality alignment, but because of external factors, which creates noise in the correlation matrix.

The decision for the number of unique interfaces and user types was primarily due to constraints in the time domain. If there was more time available, a more thorough study could be performed. This could entail creating a broader spectrum of interfaces to target niches that were not as focused on. For instance, this study only targets traditional scrolling feed social media. Idiosyncrasies of purely chat based ones such as WhatsApp, Viber or Messenger could be targetted by creating unique interfaces, thereby broadening the scope of the study and gathering more useful data.

Something that might be useful to consider further might be the effect that seeing even vaguely familiar interfaces has on users. Indeed, it might be that the user prefers a given interface not because they enjoy its traits but because they associate the general aesthetic with that of a real system (Facebook for B, Instagram for C). It was moderately remedied by mixing elements of different social media systems to ensure this does not happen. Nevertheless, this is somewhat inevitable since, as aforementioned, the elements of existing social media systems are likely chosen due to their intended effects on the user or for creating a specific atmosphere. Thus by a process of convergent evolution of these variables, the same conclusions are occasionally reached when designing the interfaces for this study. If this study were to be repeated, this could be fixed by utilising a broader range of interfaces, since they would no longer rely

as heavily on one design language.

Regarding the questionnaire, the time spent on social media is a self reported measure and thus might not be as accurate since it is a statistic people might not be inclined to disclose. Furthermore, the limited TIPI instrument was only chosen due to the time constraints for the questionnaire. As it was designed to be a very brief version of the standard tool, using its full version might have granted a more complete view of the data.

With only two items per trait for TIPI, if a participant misreads one question, the entire score for that trait is affected. Such errors are averaged out across more items in longer and more involved instruments. Assessments like the BFI-44 or NEO-PI-R break the traits into more subcategories, giving the study more granularity. For instance, Conscientiousness down into Orderliness, Productivity, and Responsibility. A full scale like this could have been used to further analyse whether a user liked an interface because they are Orderly (they like the compartmentalization) or because they are Dutiful (they wanted to finish the task correctly).

The usability is also self reported. This could be remedied in a future study by using Eye-Tracking Glasses (ETG) to track the area of interest as the interface is interacted with. This would allow for more empirical data to be collected and analysed, to act as proxies for cognitive load, attention or comfort with the UI, thereby potentially allowing for stronger conclusions. For example, changes in the diameter of the eye's pupil are used to estimate cognitive load. A 2017 study [73] found that participants using interfaces designed for their specific personality type exhibited smaller pupil diameters, indicating a lower cognitive load and an effortless experience. The amount of time eyes remain still on a specific part of the UI further measures attention. Though this would require expanded ethical approval, it would generate more objective data regarding the effect of the interface on the user.

Paid participants are also often motivated by the financial reward rather than interest in the research. This might lead to an effect where a participant provides good enough answers to get paid without putting in unnecessary effort. A 2014 meta-analysis by Cerasoli et al. [13] further supports this, showing that while extrinsic incentives can boost the performance, they can also decrease the quality of responses, especially on tasks requiring a high cognitive load. Participants may rush through the UI to maximize their hourly rate, thus failing to be fully immersed in the scenario. Another danger is users choosing the same response (e.g. "Neutral") for every question in a grid, to speed through the survey as quickly as possible due to financial motivations. Because of this, the UEQ answers were scored for hypocritical answers as outlined by its documentation [40]. No participants' results were left out from the study due to failing this. In addition, an attention checking question was present in the questionnaire. It was answered correctly by every single participant.

Nevertheless, research [28] shows that while participants start with high attention, their efforts drops towards the end of a long survey as users are more likely to give faster, shorter, and more uniform answers. As surveys also get longer, response times per question decrease. In fact, even the stated length of a survey before it is actually taken can trigger anticipatory fatigue and lower data quality from the very first question.

Chapter 6

Conclusion

The findings of the study to an extent support the idea outlined in the background section. That is, developers should move beyond efficiency as the only metric when developing an interface. As the findings showed, something that all users value is beauty in the way the interface present itself. This is an observation of the halo effect [54], that being a cognitive bias where a positive trait of a product (In this case high perceived beauty) disproportionately causes an overall positive judgment of it. The users with a higher Positive Affect tended to select beauty as a perceived trait of their tested interfaces, indicating a relationship is present. This is already something leveraged by lots of companies, since clear styles exist of UI which shift according to general preferences of the population [77]. Users judge a UI's beauty based on how well it fits the artistic trend of the current era. The more beautiful looking products also in general tend to be better rated by users [55]. However there are dangers to overlooking other factors in favour of it. For instance, the iOS 26 visual update caused a split in the user base, with some preferring the new aesthetic with a reduction in usability, whilst others criticized the changes [11]. This underlines that beauty should not be considered as the sole factor, pointing to the importance of putting usability above a pleasing design.

An additional real world implication of this research is the possibility of implementing personality-based versions of the same system to be used for different people. Akin to a dark / light mode, this would give further power to the users by letting them select the version that aligns with their needs. Though such features are somewhat present in software such as IDEs, which often include the ability to change how windows are structured, they do not let the user fundamentally change how information is presented for them. Combining this study's findings with other research [4], some implementations can be suggested for interfaces wishing to target users higher in the Conscientiousness and Agreeableness.

Instead of a fixed feed, they might allow users to switch between a List View (high density), Gallery View (visual), or a Timeline/Chronological View (logical order). This would satisfy the need for categorization. More robust search filters are also in line with such a paradigm (e.g., Show only [Educational] content from [Following] that is [Unread]). Completeness is also rewarding so design elements like "You are all caught up" provide the structured experience they enjoy. In addition, strategies can also be suggested to encourage interaction from those who might score lower on the Extraversion scale [86], [56]. Primarily, the perceived social cost of engagement must be reduced. For instance, instead of prioritizing a text comment in the selection hierarchy, using simple sticker reactions might be more effective to encourage interaction. Less open ended prompts for post composition such as "What is one thing you learned today?" further give the user a less open ended instruction, thus removing perceived friction involved in creative thinking. Using avatars instead of real photos can further push the system away from emulating a real life interaction, making users more likely to post their opinions as they feel less social pressure. The clear visibility of community rules and moderators also creates explicit, omnipresent social norms which motivate such encourage such users to participate.

Whilst a mode for users high in Extraversion and low Neuroticism [4] would include features to cater to reducing barriers between the users and the social interaction. As seen with the camera front page, such users perhaps do not want significant friction in the interface to deal with so a solution might be to remove preview screens, allowing the users to instantly capture and post their content. Such users could also get more open ended prompts for posting, including AI-generated ideas for content. Moreover, information bubbles with persistent, high-visibility indicators of which people are currently online and interacting with for a post would also encourage such people to interact more, since they thrive on participating in explicitly social activities. Bright colours that increase visual energy such as red-orange

with vibrant saturated hues are also preferred. Lastly, Extraverts are capable of scanning and retrieving information quickly in exploratory tasks. A higher information density that provides a global view with more windows and functionalities simultaneously can help them build a mental model faster than a switching-style interface. An instance of this was the multitasking capacity of interface C, with messages, feed and stories all positioned on the screen at the same time. Extraverts typically prefer non-linear navigation paths, allowing them to make their own global connections across the system rather than following a rigid, sequential path.

Another factor that is worth further elaboration is the high novelty that highly satisfied UI C users are more likely to observe. At the same time, such users have a reduced appreciation for high perspicuity traits. As outlined in [72], introducing small micro boundaries in social media actually brings about more memorable sessions and satisfied users. It could be argued that some of such micro boundaries fall under the high perspicuity UEQ trait. For example, stopping cues like Instagram's "You're all caught up" or LinkedIn's "That's all the news for now," are clear, logical conclusions to a task (scrolling). It transforms an infinite, feed into a structured, finite experience. As the study above found, these features do make the user more frustrated since their experience is interrupted. But the features that introduce friction nevertheless make the experience more memorable for the users by discouraging passive, mindless scrolling. Even though such features are supported by research to be beneficial for the average user, that very same user would likely not choose to have them implemented and their immediate emotional response is one of frustration.

This highlights the dilemma developers are faced with. Should they implement such features and risk losing users or prioritize the positive outcomes for those who wish to stay? An example is the social media app BeReal. In a traditional feed (as in TikTok), a user opens the app and content is instantly served to them. BeReal however implemented a mandatory interaction Microboundary: you cannot see what your friends have posted until after you have posted your own photo for the day. Though this app does not directly compete with the likes of Instagram and TikTok, it nevertheless occupies a sizeable niche space in the market. After hitting a peak user count in 2022, it has gone down significantly. This might partially be due to the barrier of entry being too high due to the Microboundaries present after the initial viral nature of it wore off, and thus users left.

However, Platforms that implement these Microboundaries first as pro-wellbeing features often gain a competitive advantage in trust. In Generation Z trust rating scores, BeReal leads at 30%, compared to Instagram (22%), TikTok (18%) and Snapchat (15%) [14]. This highlights an advantage of choosing this approach, which could serve as a motivation for companies to implement such features.

Conversely, other platforms may deliberately aim to exploit some user's tendency towards mindless scrolling habits. If users high in Openness and those seeking high stimulation (UI C) are more drawn to novel, less-structured interfaces (ones with less Microboundaries), they may be more at risk for addictive usage patterns. A 2025 meta-analysis [48] identified three distinct user profiles for social media addiction.

The researchers found that Cluster 2 users, those who are defined by High Openness, High Extraversion, and Low Conscientiousness, scored the highest across all social media addiction metrics. Because these users are both socially driven and curious for new experiences, they are uniquely vulnerable to interfaces that promise an infinite amount of new things to see. This is well characterized by TikTok or Instagram reels. It features the infinite scroll, has its algorithms tuned for novel content to keep the users attention and allows for a passive browsing experience with an autoplay function. A way to curb this could arguably be to create a user personality profile as a part of setting up of the account. This data would then be used to tailor the feed so that the more novel and stimulating design could be kept, with smaller additional Microboundaries, to require the users interaction. Otherwise, a more Agreeable and Perspicuity geared user might have a UI tailored with a logical structure and organised interest tabs. This might mitigate mindless scrolling and improve the user's subjective satisfaction.

There are some disadvantages to this however. The data for the users' personality profile would need to be gathered somehow, which might provide a further barrier. To make sure the data is valid and can be actually used for in depth personality profiling it would need to be from a longer survey. In addition, there might be data privacy concerns from the users since this would represent additional personal information a company might have to store. There is also a high danger of severe consequences in the case of misuse of this information, such as for targeted adverts or data leaks. Strong policy would need to be introduced to tackle this.

More research would need to be done on the kind of balance a typical user would find acceptable for data privacy and harm prevention, to determine whether this is a realistic approach

6.1 Further Research

Another notable link is the relationship between the Openness and Agreeability trait and an overall higher Positive Affect score. This highlights that some users' personalities simply predispose them to enjoying such interfaces overall, making it more difficult to identify a pattern. Future research might consider utilizing a baseline affect before the participant has performed any tasks. As a result, each participant would now have their own emotional control, which could be used more efficiently to determine any changes in their affect even if they are likely to respond positively overall.

As mentioned, a further study to more accurately measure the level of satisfaction a users exhibits when they are interacting with an interface could be done via eye-tracking. A future study could investigate to what extent a system makes a user feel or behave a certain way, or whether users of such predispositions simply tend to gravitate towards that system. This could perhaps be done via the Longitudinal Cross-Lagged design. The same group of participants would be tracked over a longer period of time. As users utilized a given system, they would be surveyed on their own perceived traits as well as their Positive Affect score. The cross lagged correlation between the usage of a system and change in a trait could then be analysed, with the following methodology implemented: If usage of UI C at Time 1 predicts an increase in Openness at Time 2, that might be Socialization occurring(the system changed the user). Conversely, if High Openness at Time 1 predicts increased usage of UI C by Time 2, that could be Selection (the user was attracted to the system).

Such a study [**?**] was performed to investigate whether social media makes people more social, or if social people just use it more. A multi year longitudinal study tracked adolescents from Grade 9 to Grade 12, measuring social anxiety and friendship quality along the way. It was found that specifically for adolescent boys, the socialization effect does take place. Socially anxious boys who used a computer to communicate with friends saw an increase in their offline friendship quality over time. The system taught them or gave them a safe space to practice social skills, which then influenced their real lives. This underlines the potential of further investigating such an effect in the personality-UI preference sphere, to gain valuable insight into the causal relationship.

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Appendix A

Appendix A: AI Prompts/Tools

The Gemini LLM was consulted about the kinds of topics, words and phrases that one might encounter on the social media inspiration was drawn from. For instance, a prompt would be asked such as "Give me some examples of typical post titles and comments on Instagram of popular or viral posts" or "What kinds of topics would the posts of a typical user's feed page be about". This was done in order to minimize the personal biases of the author, when crafting the content in the interface. Creating content that is similar to what the user likely sees on the real-life system is more likely to make them feel immersed in the tasks and thus give accurate answers. AI is especially useful for this tasks since it likely has been trained on the contents of these websites and the LLM is thus likely very accurate in giving a general idea of what a typical Instagram feed looks like.

Additionally, AI was used in a limited fashion to generate the code to create and stylize the graphs seen in the Data Analysis chapter. Prompts such as "Create a framework for a radar chart with three layers of data (blue, orange and green) with text size ..." were used. The advantage of such an approach was a more convenient graph designing process, ensuring the design stayed consistent across the various figures generated and allowing for more time to be spent on the analysis section.

The Gemini LLM was also used to generate simple boilerplate code for an initial baseline mock interface. Prompts such as "Generate a baseline layout with blank elements for a social media application using React, TypeScript, and Tailwind CSS. Create a layout component with a navigation bar, a main content area for a feed, a sidebar and a space for an account page ..." were used. This was to create a simple framework that all mock interfaces shared, so that more time could be spent on tailoring each interface to the specified parameters and exploring design ideas than manual implementation of redundant structural components.